

Associative Recurrent State vs. Key-Value Attention in In-Context Demonstration Learning

Adhithya Navaneethakrishnan | DataForge 2026: Pathway Track (NeurIPS 2026 Education Format)

Artifact URL: <https://bdh-memory-explainer-nexora-7367.vercel.app/>

Source Code: <https://github.com/Adhi426/bdh-memory-explainer>

1. Problem Definition & Architectural Motivation

Standard Transformers process in-context demonstrations (e.g., few-shot prompt exemplars, multi-turn reasoning traces, or visual transformation pairs) by storing the projection tokens in an explicit Key-Value (KV) cache. For a sequence of N demonstration tokens with embedding dimension d , memory scales as $O(N \cdot d)$. At long sequence horizons, this KV footprint induces high-bandwidth memory (HBM) bandwidth saturation, restricts batch throughput, and enforces quadratic attention compute during token generation.

Pathway's Dragon Hatchling (BDH) and BDH-CQ architectures reformulate contextual memory by treating attention not as a growing cache, but as recurrent synaptic plasticity. Rather than preserving every token vector in high-bandwidth memory, demonstrations are accumulated into a fixed-shape associative state, providing unbounded sequence processing in $O(1)$ memory without dynamic allocation bottlenecks.

2. Theoretical Mechanism: Additive Synaptic Memory & Latent Recurrence

BDH-CQ replaces dynamic token caches with an associative recurrent memory matrix $W \in \mathbb{R}^{d \times d}$ [cite: 3]. For each contextual demonstration pair (x_t, y_t) with key projection $k_t = f(x_t)$ and value projection $v_t = g(y_t)$, memory writes occur via an additive Hebbian formulation [cite: 3]:

$$W_t = (1 - \gamma)W_{t-1} + \eta(k_t v_t^T)$$

where $\gamma \in [0, 1)$ represents synaptic decay (recency discount) and η is an update scaling factor [cite: 3]. Context retrieval for an unseen test query $q = f(x_{\text{test}})$ is computed as a direct linear projection [cite: 3]:

$$\hat{v} = qW_t$$

Unlike token-generating autoregressive models that verbalize intermediate thoughts through natural language Chain-of-Thought (CoT), BDH-CQ performs inference-time scaling through **recurrent latent-space reasoning** [cite: 3]. The internal hidden state h is iteratively refined across T recurrent workspace passes [cite: 3]:

$$h^{(\tau+1)} = \sigma \left(h^{(\tau)} W_t + b \right)$$

where σ applies a non-negative thresholding activation enforcing biological sparsity [cite: 3].

3. Architectural Comparison

Dimension	Standard Transformer	State-Space Model (Mamba)	BDH-CQ (Pathway)
Context Memory Footprint	$O(N \cdot d)$ (Linearly expanding)[cite: 3]	$O(S)$ (Compressed hidden state)[cite: 3]	$O(d^2)$ (Fixed Synaptic Weight Matrix)[cite: 3]
Context Accumulation Rule	Dynamic concatenation[cite: 3]	Continuous-time linear ODE / Scan[cite: 3]	Additive Hebbian outer product writes[cite: 3]
Reasoning Substrate	Serialized verbal tokens (CoT)[cite: 3]	Token-by-token recurrence[cite: 3]	Latent workspace recurrent passes[cite: 3]
Task Efficiency (ARC-AGI)	\$0.015 – \$0.040 / task	Variable / Unverified[cite: 3]	\$0.0007 / task (Reported pass@2)[cite: 3]
State Nature	Non-parametric growing cache[cite: 3]	Evolving parametric vector[cite: 3]	Synaptic fast weights / Associative matrix[cite: 3]

4. Empirical Benchmark Evidence (ARC-AGI-1)

On the Abstraction and Reasoning Corpus (ARC-AGI-1)—which evaluates a system's ability to infer spatial transformation rules from sparse demonstrations—a 150M parameter BDH-CQ model achieved **29.5% pass@2** at an inference cost of **\$0.0007 per task** without verbalized chain-of-thought[cite: 3]. By scaling compute along the latent recurrent dimension rather than generating verbose natural language tokens, BDH-CQ operates on a superior cost-accuracy Pareto frontier compared to frontier LLM prompting frameworks[cite: 3].

5. Falsifiable Limitation & Failure Boundary

While fixed associative recurrent memory resolves the $O(N)$ KV memory wall, it introduces an inherent trade-off: **associative cross-talk under cue collision**[cite: 3]. Linear retrieval $\hat{v} = qW_t$ requires keys $\{k_1, \dots, k_N\}$ to remain approximately orthogonal[cite: 3]. When demonstration tasks contain collinear cues or overlapping low-rank subspaces ($k_i^T k_j \rightarrow 1$), the outer-product terms in W_t interfere constructively and destructively[cite: 3]. While softmax attention isolates overlapping tokens through non-linear exponentiated competition, BDH-CQ's linear readout experiences representation blurring and corrupted predictions when cue similarity exceeds the network's capacity[cite: 3].

6. Primary References

1. Kosowski, A., et al. (2025). *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*. arXiv:2509.26507[cite: 3].

2. Pathway Research. (2026). *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*. arXiv:2608.09888[cite: 3].

3. Vaswani, A., et al. (2017). *Attention Is All You Need*. Advances in Neural Information Processing Systems (NeurIPS)[cite: 3].